

Chapter 84 — Inguinal Node Dissection

- **MacCormac (1886)** described the first penectomy series; **Basset (1912)** first described **ILND** (inguinal lymph node dissection) for cancer, as en bloc removal of the superficial and deep packets. **Daseler (1948)** described **radical ILND** and the five-zone nodal anatomy.
- **The inguinal nodes receive metastases from penile, vulval and rectal cancer and melanoma.**
- **Penile cancer is rare, so there is no level I evidence;** practice rests on small series and retrospective data.

CURRENT REAL-WORLD PRACTICE

- **Adherence to ILND guidelines is poor** in United States registries (Surveillance, Epidemiology, and End Results [SEER]; National Cancer Database), the Swedish national register and multicentre European-American series. Even patients at high risk by European Association of Urology (EAU) criteria are often not dissected.
- **An adequate ILND removes at least 8 nodes per side,** yet few dissections reach this.
- **Few clinicians treat many cases a year, and practice varies widely by region.** In intermediate-risk disease with impalpable nodes, a substantial minority chose surveillance. Invasive staging was split between dynamic sentinel node biopsy (DSNB), modified ILND, superficial ILND and video-endoscopic inguinal lymphadenectomy (VEIL). For node-positive disease, **adjuvant chemotherapy was preferred to neoadjuvant, except in cN3 disease.**
- **Staging is often omitted in pT1b–T3, clinically node-negative (cN0) disease.** Omission is associated with older age, comorbidity, lower stage, Medicaid insurance and non-academic care, and **independently predicts overall mortality.**
- **Guideline-concordant nodal management is associated with better overall survival,** so fear of morbidity should not lead to omission.

Current Indications and Guidelines

- **There is no American Urological Association (AUA) guideline.** The EAU–American Society of Clinical Oncology (EAU-ASCO, 2023) and National Comprehensive Cancer Network (NCCN, 2023) guidelines each use three risk tiers, defined differently because level I data are lacking (Table 84.1).
- **Risk grouping has high false-positive and false-negative rates.** Occult nodal metastasis is uncommon in low-risk disease and substantial in intermediate- and high-risk disease.
- **Low risk with impalpable nodes: surveillance without routine staging imaging,** because occult metastasis is unlikely.
- **Palpable nodes or high-risk disease: fluorine-18 fluorodeoxyglucose positron emission tomography-computed tomography (18FDG-PET/CT), or computed tomography (CT) of the chest, abdomen and pelvis, before treatment.**
- **Consensus: complete inguinal and pelvic management within 3 months of diagnosis,** unless upfront chemotherapy is given.
- **Grading is poorly reproducible between pathologists,** which limits prognostication. Cases should have **expert genitourinary pathology review,** and patients should be counselled about under- or over-staging.

TABLE 84.1 RISK STRATIFICATION AND TREATMENT APPROACHES BASED ON DIFFERENT GUIDELINES

	Low risk	Intermediate risk	High risk
NCCN definition	pTis, pTa, pT1a	pT1b, G1–2	pT2 or higher, or G3/G4
		G2 pT1a	pT1b or higher

	Low risk	Intermediate risk	High risk
EAU-ASCO definition	pTa, pTis, G1 pT1a		
NCCN, impalpable nodes	Surveillance	Surgical staging: modified or radical bilateral ILND, or DSNB	Same as intermediate
EAU-ASCO, impalpable nodes	Surveillance	Surveillance versus surgical staging , case by case (DSNB, or ILND if DSNB cannot be offered)	DSNB , or ILND if DSNB cannot be offered

G = pathological grade.

- **What the table tests:** the guidelines agree on low risk but **draw the intermediate boundary one step apart**. Under NCCN, pT1a is always low risk and pT1b is intermediate unless G3/G4. Under EAU-ASCO, pT1a G2 is already intermediate, and **every pT1b is high risk**.
- **Trap:** a pT1b G2 tumour is intermediate risk under NCCN but high risk under EAU-ASCO.
- EAU-ASCO offers surveillance at intermediate risk only to patients willing to comply with strict follow-up, weighing the risk of metastasis against the morbidity of staging.

NCCN VERSUS EAU-ASCO: DECISIONS BEYOND TABLE 84.1

Decision	NCCN	EAU-ASCO
Staging tool, impalpable high-risk nodes	ILND; DSNB if the clinician is experienced	DSNB first ; ILND only if DSNB is unavailable and referral is not feasible, or the informed patient prefers it
Palpable nodes	Percutaneous biopsy → excisional biopsy if negative → surveillance if both negative; either positive → bilateral ILND , with upfront chemotherapy if eligible	Not detailed separately in the chapter
Bulky or fixed nodes	Both: upfront cisplatin- and taxane-based chemotherapy in patients who can tolerate it	

Concept — The guidelines agree at the extremes and diverge in the middle. Both survey low risk and stage high risk. They differ in **where the intermediate boundary falls and which staging tool comes first**: NCCN leads with dissection, EAU-ASCO with sentinel biopsy.

BENEFITS OF INGUINAL LYMPH NODE DISSECTION

- **Penile cancer usually has a contained locoregional phase in which inguinal disease precedes systemic spread.** In many other cancers, nodal disease signals systemic disease; here it creates a **window of cure**.

EARLY VERSUS DELAYED ILND (Kroon et al., 2005; non-randomised, T2–3, impalpable nodes)

Early (upfront) ILND	Delayed ILND (when nodes become palpable)
Extranodal extension (ENE) about 20%	ENE about 95%
3-year disease-specific survival about 84%	3-year disease-specific survival about 35%

- **ILND should therefore not be delayed beyond 3 months** in high-risk surgical candidates with impalpable nodes.
- **The presence and extent of nodal involvement is the strongest prognostic factor:** five-year cancer-specific survival (CSS) falls steeply from N0 and N1 to N2 and N3. **ENE, even in a single node, is classified as pN3.**

- **Nodal yield is the surrogate for oncological adequacy.** Removing 8 or more nodes is associated with better survival in grade 3 or pT2–4 disease. There is no universal cut-off, but surgeons should **aim for at least 8 nodes per side** when safe, accepting that higher yield brings more complications.

Concept — *In penile cancer the groin is a curable station, not a marker of systemic disease. This is why **timing (within 3 months) and completeness (at least 8 nodes per side)** are oncological variables, not technical refinements.*

PATIENT SELECTION

- **Human papillomavirus (HPV)-positive tumours have better disease-specific survival on meta-analysis, but the nodal data conflict.** Some series show less lymphatic tumour embolisation, lower clinical N stage and lower nodal metastasis density with HPV positivity. Others show no difference in N classification, disease-specific or overall survival, or upstaging from cN1 to pN2/N3 at ILND. **HPV status is therefore not reliable for choosing ILND or surveillance.**
- **About half of palpable nodes, and up to a quarter of impalpable groins, harbour metastasis,** so surgery carries a real risk of overtreatment.
- **Nomograms predicting nodal metastasis exist, but, like HPV status, are not recommended for routine use,** especially where they produce guideline-discordant advice.
- **The guidelines select by primary-tumour pathology:** T stage, grade, lymphovascular invasion, perineural invasion and sarcomatoid features.

Concept — *Patients are selected on primary-tumour pathology alone. HPV status and nomograms are prognostically interesting but are not decision tools: their nodal predictive value is inconsistent, and they can pull recommendations away from the guidelines.*

LYMPHATIC DRAINAGE

SUPERFICIAL VERSUS DEEP PENILE LYMPHATIC SYSTEMS

Superficial system	Deep system
Drains penile shaft skin and prepuce	Drains glans and frenulum
Dorsal capillary network → divides at the penile base into right and left branches → ipsilateral inguinal nodes	Trunks encircle the corona , merge dorsally, run within Buck's fascia → presymphyseal lymphatics → nodes of the femoral triangle

- **Lymphatics cross over in the presymphyseal region, so contralateral spread is possible;** this is the basis for bilateral dissection. **With unilateral palpable nodes, the contralateral groin harbours micrometastasis in about 30%.**
- **After late inguinal recurrence following a long latency, views differ** on whether unilateral ILND suffices or bilateral dissection is needed.
- **Spread is stepwise: superficial inguinal → ipsilateral deep inguinal → external iliac nodes.** The side of drainage **cannot be predicted from the site of the lesion** on the penis.
- **As a rule, drainage does not skip.** The rare reports of positive iliac nodes with negative inguinal nodes are **probably due to undersampling at ILND or inadequate pathology review.**

Concept — *Two rules generate the surgical logic. Drainage is **unpredictable in side but orderly in sequence.** The first rule demands bilateral staging. The second allows superficial status to gate deep dissection, and inguinal status to gate pelvic dissection.*

PREOPERATIVE WORKUP

Physical Examination

- **Examine the groins at the first evaluation** and classify as **cN0 (impalpable) or cN+ (palpable)**. For palpable nodes, record number, location, laterality, and whether mobile or fixed.
- **If examination is indeterminate** (obesity, previous groin surgery), obtain magnetic resonance imaging (MRI), CT or PET/CT. Centres with melanoma experience also use groin ultrasonography (US).
- **Infection of the primary can cause reactive nodes, but a trial of antibiotics with observation is not recommended**, because it delays diagnosis and treatment.

Cross-Sectional and Nuclear Imaging

- **CT of the chest, abdomen and pelvis detects distant metastasis** in patients with suspicious nodes or a high-risk primary.
- **For the groin itself, imaging misses deposits below 1 cm**, so it is reserved for when examination is unreliable (obesity, previous surgery or radiotherapy). FDG-PET is therefore unlikely to change management in the impalpable groin (see the modality table below).

Ultrasonography, Fine-Needle Aspiration Cytology and Dynamic Sentinel Lymph Node Biopsy

- **An involved node on US** shows altered architecture and is hypoechoic and heterogeneous.
- **Fine-needle aspiration cytology (FNAC), guided by US or CT, is specific but insensitive**, whatever the guidance. **Positive → ILND; negative → DSNB or ILND**, never discharge.
- **US is performed before or at DSNB** to find and sample enlarged nodes, for this reason:
 - A node replaced by tumour **blocks lymphatic inflow**, and drainage is **rerouted to a neo-sentinel node**.
 - The blocked node takes up **no radiotracer or dye**, so DSNB is **falsely negative**.
 - Adding US **improves DSNB accuracy and lowers the false-negative rate**.
- **Single-photon emission computed tomography-CT (SPECT-CT) in palpable groins** shows frequent rerouting to neo-sentinel nodes, occasionally contralateral, and sometimes absent drainage.

DSNB BY SITE OF THE PRIMARY

Prepuce or shaft skin lesions	Glans lesions
DSNB is useful	Caution: the glans can drain directly to a deep inguinal node , so sampling only the superficial superomedial nodes can give a false negative

- **Historical false negatives were partly due to pathological undersampling**; serial sectioning of the whole node at **150 micrometres** reduces this.
- **Sequence: US ± FNAC → if negative → DSNB with lymphoscintigraphy using technetium-99m (99mTc) nanocolloid plus blue dye**.
- **DSNB has high sensitivity and very high specificity on systematic review**, but results depend on the surgeon; **both guidelines require experienced hands**.
- **A negative DSNB does not exclude later nodal disease**, so patients enter surveillance (schedules under Follow-Up).

STAGING MODALITIES FOR THE GROIN COMPARED

Modality	Role	Weakness
Physical examination	First step: classifies cN0 or cN+	Unreliable in obesity or after surgery; about half of enlarged nodes are reactive
CT, MRI, PET/CT		Insensitive below 1 cm

Modality	Role	Weakness
	Distant staging; groin assessment only when examination fails	
FDG-PET/CT	Palpable nodes and distant staging	Sensitive in palpable nodes; poorly sensitive in impalpable nodes
Lymphotropic nanoparticle-enhanced MRI (LNMRI)	Investigational; promising early sensitivity	Unreliable for micrometastasis (2 mm or smaller)
US	Detects abnormal nodes; guides FNAC; precedes DSNB	A negative scan does not exclude metastasis
FNAC	Confirms metastasis: positive → ILND	Low sensitivity: negative → DSNB or ILND
DSNB (99mTc plus blue dye)	Minimally invasive staging of impalpable nodes	Operator-dependent; false negatives from blocked nodes and glans-to-deep drainage

Concept — Every groin staging test fails in the same direction: false negatives from small or tumour-replaced deposits. The workup is therefore a funnel in which **a negative result never ends the pathway**: negative FNAC leads to DSNB, and negative DSNB leads to surveillance.

KEY POINTS

- Lymph node metastasis is the strongest prognostic factor for survival in penile cancer.
- Real-world adherence to guideline recommendations on the inguinal nodes is low.
- Address the inguinal nodes within 3 months of diagnosis; delay worsens oncological outcomes.

SURGICAL APPROACH

Anatomy

Camper's and Scarpa's Fascia and the Arterial Supply to the Skin

- Skin flaps are 5–7 mm thick, raised deep to Camper's fascia along the plane of Scarpa's fascia. Camper's fascia carries a rich capillary network, so staying beneath it **minimises flap necrosis**.
- The subdermal supply runs lateral to medial along Langer's lines. An incision parallel to these lines preserves skin viability; one that crosses them transects the vessels.

FLAP THICKNESS: THE TWO FAILURE MODES

Flap too thin	Flap too thick
Breaches Camper's capillary network → flap necrosis	Leaves superficial nodes adherent to Scarpa's fascia → inadequate resection

Superficial and Deep Nodal Packets

- The superficial packet lies above the fascia lata and holds 4–25 nodes, in five zones.

DASELER'S FIVE ZONES OF THE SUPERFICIAL INGUINAL NODES (*Daseler et al., 1948*)

Zone	Related vessel
Central	Saphenofemoral junction
Superolateral	Superficial circumflex iliac vein
Inferolateral	Lateral femoral cutaneous and superficial circumflex veins
Superomedial	Superficial external pudendal and superficial epigastric veins

Zone	Related vessel
Inferomedial	Great saphenous vein

- **Sentinel nodes lie mainly in the superomedial zone, then the central zone.** Spread to the two inferior zones without sentinel involvement has not been reported.
- **The deep group has 1–3 nodes, medial to the femoral vein in the femoral canal.** The **node of Cloquet**, the most cephalad, lies **between the femoral vein and the lacunar ligament** and is reachable from both inguinal and pelvic approaches.
- **Pelvic spread usually requires ipsilateral inguinal metastasis first;** crossover from one groin to the contralateral pelvis is **extremely rare**.

Concept — *The superomedial sentinel pattern and the no-skip sequence justify every reduced template. Tissue that is never the first station can be left behind safely **when the first station is clear**.*

Landmarks and Templates

- **Standard radical ILND boundaries (the femoral triangle):** inguinal ligament superiorly, adductor longus medially, sartorius laterally, and the crossing of these muscles inferiorly.
- **Positive deep nodes have been found only with positive superficial nodes**, so some experts omit deep ILND when the superficial nodes are negative.

DISSECTION TEMPLATES COMPARED

Template	Extent	Key features
Standard radical	Whole femoral triangle, superficial and deep	Fascia lata opened; saphenous vein often ligated; sartorius transposed
Superficial (fascia-preserving)	Nodes above the fascia lata only	Fascia lata preserved
Deep	Fascia lata resected; nodes beneath it, medial to the femoral vessels	Added when the superficial nodes are positive
Modified (Catalona)	Excludes the area lateral to the femoral artery and caudal to the fossa ovalis	Smaller incision, saphenous vein preserved , thicker flaps, no sartorius transposition
Limited	Cranial border from anterior superior iliac spine to pubic tubercle; perpendiculars of 20 cm laterally and 18 cm medially ; caudal line joining their ends	No recurrences in node-negative patients on long follow-up, despite substantial wound morbidity

- **Modified and limited templates cause less morbidity than standard radical ILND;** the very high historical morbidity fell at least partly through template modification. They appear **oncologically acceptable in selected patients with a negative frozen section**.

Concept — *Every reduced template trades coverage of low-yield territory (lateral, inferior, deep) for **preserved flap blood supply and lymphatics**. The trade is safe only because of the sentinel map and the no-skip rule.*

Saphena Magna and Fossa Ovalis

- **The great saphenous vein, the longest vein in the body,** drains the foot and lower limb, passes behind the medial condyle of the femur, and **enters the femoral vein through the fossa ovalis**, an oval opening in the fascia lata just caudal to the medial inguinal ligament.
- **The deep nodes lie medial to the saphenofemoral junction.** Radical ILND opens the fascia lata, so the vein is sometimes ligated at **two sites: caudally at the apex of the femoral triangle, and cephalad at the saphenofemoral junction**.

Relationship to Nearby Structures

- The spermatic cord crosses medial and cephalad to the fossa ovalis, over the inguinal ligament. With bulky nodes, **identify and medialise the cord early**.
- The femoral nerve, the largest branch of the lumbar plexus (**dorsal divisions of the ventral rami of L2–L4**), runs deep to the iliacus fascia and **under the inguinal ligament just lateral to the femoral artery**. It is motor to pectineus, quadriceps femoris and sartorius, and sensory to the anterior thigh.
- The anterior cutaneous branches to the groin skin are commonly sacrificed, but the motor nerve must be preserved: **never dissect lateral to the femoral artery**.
- Fig. 84.1 shows the modified field lying inside the standard field, sparing the lateral and inferior territory, with the superficial tributaries marking Daseler's zones. Fig. 84.3 shows the order at the fossa ovalis, lateral to medial: **nerve, artery, vein**.

Concept — The femoral artery is the **lateral boundary of safety**. The motor nerve lies beyond it, the sentinel nodes lie medial to it, and the modified template stops at it.

SURGICAL RESECTION

Open Approach

Sterility

- The groin is colonised in skin folds and hair, and wound infection or infected lymphocele severely impedes recovery. **Shave carefully, prepare thoroughly, drape appropriately, and apply a sterile dressing at the start.**

Incision and Dissection

- **Position supine in frog-leg** (thigh abducted and externally rotated, knee flexed and supported); in obesity, **tape the pannus to the chest**. The anaesthetist may be asked to keep the patient **incompletely paralysed**.
- **Sequence:** mark the borders → **oblique incision about 2 cm below and parallel to the inguinal ligament**, along Langer's lines, across the width of the dissection → **flaps 3 cm cephalad to the ligament and caudally to the apex** → sweep fat and areolar tissue off the external oblique aponeurosis and cord down to the ligament (superior boundary) → at the apex, **identify and usually divide the great saphenous vein** (spare it if metastatic disease is minimal) → incise the fascia lata over sartorius laterally and adductor longus medially → identify the femoral artery and vein at the apex and **dissect cephalad along their anterior surfaces**, ligating cutaneous perforators → **divide the saphenous vein again at the saphenofemoral junction** → clear the deep nodes up to the femoral canal → **never expose the lateral surface of the femoral artery** (femoral nerve, deep femoral artery).
- **Reconstruction:** detach sartorius from the anterior superior iliac spine and **transpose it, or roll it 180 degrees medially, over the femoral vessels (Baronofsky manoeuvre)**, suturing it to the inguinal ligament and adjacent muscles → **close the femoral canal by suturing the shelving edge of the inguinal ligament to Cooper's ligament**, without narrowing the external iliac vein or injuring the inferior epigastric vessels → **closed-suction drains** subcutaneously → **tack the flaps to the muscle** to obliterate dead space → absorbable subcutaneous sutures or staples.

OBLIQUE INCISION VERSUS GIBSON INCISION

Oblique incision below the ligament	Large Gibson incision, dissecting inferiorly
Standard; best exposure of the full template	Less skin devascularisation (superior border intact); easy access for iliac dissection; but limited view , especially with a full template or obesity

- **Choose the incision that allows the best resection**, because ILND can be curative and resection quality affects outcome.
- **Large skin defects:** close with a **scrotal rotation flap, abdominal wall advancement flap, or rectus abdominis or tensor fasciae latae myocutaneous flap**. Where natural tissue is not available, **negative-pressure wound therapy** closes the defect in time; drainage may reach **a litre a day or more**, so oral intake must keep pace.

Use of Ultrasonic Devices versus Electrocautery for Lymph Leak

- **Lymphatics should be controlled to reduce lymphocele**, but no method is agreed: clips, ligatures, vessel-sealing devices and monopolar electrocautery are all used.

MONOPOLAR ELECTROCAUTERY VERSUS VESSEL-SEALING DEVICES

Monopolar electrocautery	Vessel-sealing devices
Thrombus formation or heat coagulation of blood in the lumen → lumen often stays patent; lymphatic closure unreliable	Protein denaturation → theoretical advantage for lymphatic closure

- **No ILND-specific evidence exists**, but sealing devices are superior in axillary, mediastinal and transplant iliac dissection. **No sealing mechanism is proven better than another.**

Does Fibrin Sealant Reduce Leakage?

- **Retrospective and non-urological data suggested less leak, but double-masked randomised trials and meta-analyses show no benefit**, so fibrin sealant is not used routinely against lymphorrhoea.

Concept — *The two major wound complications have two separate causes. Flap necrosis comes from **skin blood supply**; lymphocele comes from **lymphatic closure**. The open technique addresses the first through the flap plane and incision, and the second through energy choice and drains. Sealants have not added to this.*

CONCOMITANT INGUINAL LYMPH NODE DISSECTION WITH PNECTOMY AND/OR PELVIC LYMPH NODE DISSECTION

- **With palpable nodes, DSNB, excisional biopsy or ILND can accompany treatment of the primary**, and a positive inguinal frozen section allows pelvic lymph node dissection at the same operation.
- **Staging is acceptable, but ILND must not be delayed beyond 3 months.** Concurrent versus staged surgery is decided with the patient, weighing age, goals of care, comorbidity and surgeon judgement.

Open versus Minimally Invasive Surgery (VEIL, Robotic VEIL)

Perioperative outcomes

- **Operating time varies widely**; the minimally invasive approach takes about **2 hours, roughly 15 minutes longer per side than open surgery**. Intraoperative complications are rare in experienced hands.
- **The data are retrospective and selection-biased**, because video-endoscopic and robotic VEIL (RVEIL) are usually reserved for **non-bulky, non-fixed nodes**.

Oncologic Outcomes

- **With little long-term recurrence data, nodal yield is the surrogate**, and it is similar between approaches. In a prospective phase I study, open re-exploration after robotic ILND found robotic dissection adequate in almost all groins, with one conversion; positive nodes were removed robotically.

- **VEIL and RVEIL cannot yet be compared**; the surgeon's experience should guide the choice so that adequacy is not sacrificed. **Large randomised trials with longer follow-up are needed.**

OPEN VERSUS MINIMALLY INVASIVE ILND

Domain	Open ILND	VEIL / RVEIL
Operating time	Shorter	About 2 hours; slightly longer per side
Blood loss	Low	Low; similar
Hospital stay	Longer	Shorter
Time to drain removal	Longer	Shorter
Wound complications, dehiscence, infection, flap necrosis, major morbidity	Higher	Lower
Nodal yield; local and distant recurrence	Reference	Similar
Suitable nodes	All, including bulky, fixed or skin-involving	Non-bulky, non-fixed
Evidence	Retrospective	Retrospective, selection-biased; no level I data

Surgical Technique

POSITIONING BY PLATFORM

Platform	Positioning
VEIL	Split-leg bed, frog-leg or low lithotomy ; surgeon between the legs, assistant at the side; two teams can operate on both groins at once
RVEIL, da Vinci Si	Dorsal lithotomy , avoiding repositioning and redocking; assistant lateral to the right leg for the right side, between the legs for the left
RVEIL, da Vinci Xi	Supine frog-leg on a non-split bed ; boom rotation reaches both sides without repositioning the robot

- **Access sequence:** sterile Foley catheter on the field → **mark an inverted triangle before insufflation** (which distorts landmarks): base from anterior superior iliac spine to pubic tubercle along the ligament, sides along sartorius and adductor longus to the apex → **2-cm incision 3 cm below the apex, about 25 cm below the inguinal ligament** → identify Scarpa's fascia (white subcutaneous layer) and **finger-dissect beneath it** → two 8-mm robotic ports, finger-guided, laterally and medially → extend the space by sweeping with the endoscope, or with a 12-mm balloon trocar → **insufflate at 25 mm Hg for 10 minutes, then work at 15 mm Hg** → 0-degree 10-mm lens, plus a 10-mm assistant port between camera and working port → dock the robot at 45 degrees, contralateral to the first procedure (right side) and ipsilateral to the patient in the second (left side).
- **Instruments:** bipolar Maryland or PK forceps in the left arm; monopolar scissors or ultrasonic instruments in the right.
- **Dissection sequence:** develop the anterior space to the **inguinal ligament (transverse white fibres, the superior limit)** → keep to sartorius laterally and adductor longus medially, confirmed by their fasciae against the markings; the cord lies medially, and **reddish muscle fibres mean the plane is deep to the fascia lata** → roll the nodal tissue inward from both sides and define the caudal apex → **follow the great saphenous vein from the apex to the saphenous arch and saphenofemoral junction** → lift the packet off the fascia lata cephalad (non-dominant hand lifts, monopolar scissors advance) → free the superolateral and superomedial limits so that superficial and deep planes meet and the packet leaves the ligament → clip venous tributaries, using **femoral artery pulsation** (or intraoperative US) as the landmark → **release**

the packet from the saphenous vein, sparing it if possible, otherwise ligating at the arch with suture, stapler or clips → retrieve in a bag through the extended camera incision.

- **The frozen section decides the ipsilateral deep dissection**; meanwhile, start the contralateral space. **Deep ILND**: re-insufflate → open the fascia lata medial to the saphenous arch → dissect inferomedially around the femoral vein **to the femoral canal, until pectineus is seen**.
- **Close**: drop pressure to **5 mm Hg to confirm haemostasis**, since lymphocele and haematoma can become infected → **drain in the most dependent (caudal) part of the field**, where fluid gathers when upright → close ports.
- **Saphenous sparing aims to reduce lymphoedema**, though not every study confirms it. **Robotic and laparoscopic ILND have never been compared**; theoretical robotic advantages are ergonomics, three-dimensional vision and **indocyanine green (ICG) fluorescence** to identify lymphatics.
- **Fig. 84.5** shows the landmarks for trocar placement (anterior superior iliac spine, inguinal ligament, pubic tubercle, cord, medial border of sartorius, lateral border of adductor longus), with the camera port about 25 cm below the ligament and working ports about 8 cm to either side.

Bulky or fixed nodes (cN3)

- **cN3 nodes can invade the femoral vessels and cause catastrophic haemorrhage**. The **preferred treatment is upfront systemic therapy, then surgery**.
- **Upfront resection, curative or palliative, is considered** if the patient is ineligible for or intolerant of chemotherapy, or **vascular invasion is imminent**.
- **If the skin is involved, perform a wide open resection** with a **gracilis or vertical rectus abdominis myocutaneous flap** over the vessels, usually with plastic surgery. Without that support, **negative-pressure wound therapy** closes the wound over about **12 weeks**.

Concept — *The minimally invasive approach reproduces the open template through a remote port, keeping the same boundaries, the same apex-to-ligament sequence, and **the same frozen-section gate for deep dissection**. It gains wound outcomes and gives up bulky and skin-involving disease, which remains open surgery.*

POSTOPERATIVE CARE

- **Care depends on approach, patient factors** (obesity, smoking, poor healing) **and surgeon preference**. Some allow same-day walking and next-day discharge, especially after minimally invasive surgery; others start with **bed rest when wound breakdown or poor healing is a concern**.
- **Drains stay for weeks, until output falls below 30 mL per day**. The authors use a **19-Fr Blake drain on a suction bulb**, removed at **below 30 mL per day after mobilisation**, avoid bed rest unless the wound is at risk, and prescribe **thigh-high compression hose for at least 1 month**.

PROPHYLACTIC ANTIBIOTICS: NCCN VERSUS THE AUTHORS

NCCN	Authors
Recommends prophylaxis ; consider a suppressive oral cephalosporin (or other broad-spectrum antibiotic active against gram-positive organisms) for days to weeks	No routine prophylaxis , because supporting data are limited

- **A Society of Urologic Oncology (SUO) survey** found that most urologists give perioperative antibiotics (many stopping at 24 hours), spare the saphenous vein, and give prophylactic anticoagulation (often stopped once walking). About half obtain frozen sections, almost all use drains, a minority use incisional negative-pressure devices, and lymphoedema is managed with compression and/or lymphoedema referral.

COMPLICATIONS

- **ILND is highly morbid:** phlebitis, wound infection, dehiscence, **flap necrosis**, lymphoedema, lymphocele, neurapraxia and pulmonary embolism. Rates vary with patient factors, template and approach, and because **there is no standard reporting system**.
- **Wound-healing problems (infection, flap necrosis) are the commonest.** Prevent them with **rigorous asepsis, appropriate antibiotics, and never dissecting above Camper's fascia**. Incisional negative-pressure therapy may be considered, without robust ILND data.
- **Minimally invasive ILND has the more favourable profile** (table above). In robotic series, lymphocele, surgical-site infection, cellulitis and early or late limb oedema are similar to open, but major complications, flap necrosis and severe limb oedema are fewer. **After robotic ILND the commonest complication is lymphocele, then lymphoedema and cellulitis.**
- **Surgeon experience probably drives both oncological outcome and morbidity;** refer patients to **high-volume centres and surgeons**.

Concept — Almost every complication of ILND traces to one of two disruptions: of the **skin's blood supply** (necrosis, dehiscence, infection) or of **lymphatic channels** (lymphocele, lymphoedema). Each refinement in this chapter targets one of them: flap plane, templates, saphenous sparing, sealing devices, and minimally invasive access.

FOLLOW-UP

- **Median time to inguinal recurrence is about 1 year;** in-field recurrence after complete ILND is exceedingly rare. **Most locoregional recurrences occur within 2–3 years**, and after 5 years recurrences are local or new primaries. This supports **intensive early follow-up, reduced after 3 years**.
- **Beyond 5 years, follow-up is case by case:** late recurrence is rare, so visits can stop in patients who understand and perform self-examination. **Those unlikely to self-examine may need long-term follow-up.**

SURVEILLANCE AND FOLLOW-UP SCHEDULES COMPARED

Setting	Clinical examination	Imaging
After DSNB, intensive schedule	Every 2 months for years 1–2, every 3 months in year 3, every 6 months to year 5	Not specified
EAU-ASCO, negative DSNB or node-negative after local treatment	Every 3 months for 2 years, then every 6 months to year 5 (penis and groins)	None; no proven benefit
EAU-ASCO, node-positive after ILND	As above	CT or MRI of chest, abdomen and pelvis every 3 months for 2 years, then every 6 months to 5 years
NCCN, pN0–pN1 (examination interval also applies to pN0 after DSNB)	Every 6 months in years 1–2, annually in years 3–4	After ILND: CT abdomen and pelvis plus chest radiograph at the same intervals
NCCN, pN2–pN3 after ILND	Every 3 months in year 1, every 6 months in years 2–4	CT abdomen and pelvis plus chest radiograph at the same intervals

- **The examinable contrast: NCCN images every patient after ILND, including pN0; EAU-ASCO images only node-positive patients.**

Concept — Follow-up intensity tracks the recurrence curve: dense in years 1–2, reduced after year 3, and handed to self-examination after year 5. **The less complete the nodal staging, the denser the schedule,** which is why the post-DSNB schedule is more intensive than the post-dissection one.

KEY POINTS

- **A contained locoregional phase gives a window of cure;** when indicated, immediate ILND improves oncological and survival outcomes compared with delayed surgery.
- **Drainage goes to either or both sides and cannot be lateralised from the lesion;** it proceeds superficial inguinal → deep inguinal → external iliac and **does not skip**.
- **Morbidity can be reduced in selected patients** with DSNB and minimally invasive techniques.
- **Counsel patients appropriately, and do not avoid this life-saving operation for fear of morbidity.**